

## Assignment 1 – Advanced Machine Learning 2026

### Instructions

- In this assignment, you will explore different methods for text representation, dimensionality reduction, feature selection, modeling, and evaluation.
- Assignments are to be done in **groups of 5**.
- Use **Python** to solve the exercise.
- Deliverables:
  - The file Team-XX.pdf (where XX is your group number, e.g. Team-07.pdf) containing your report.
  - A file Team-XX-text.pdf containing the your report, excluding figures, tables, references and technology statement. This will be used to check whether you adhere to the 6 page maximum.
  - Python code that you have developed to solve the assignment. Keep the code of methods that did not make it into the final model, but do not submit code that does not work. Make sure that this code is **readable and concise**. If you developed multiple files, place them in a zip file named Team-XX-code.zip.
  - A Team-XX.pickle file of your best-performing **LogisticRegression** model (this can use either TF-IDF features or Word2Vec features). This file can be created using the pickle package in the base installation of Python. **Save the fitted model together with everything it needs in order to make a prediction**, the fitted vectorizer and/or scaler it depends on, most easily by pickling a single scikit-learn Pipeline.
  - Submit all deliverables electronically through **CANVAS**. Make sure to submit everything in a single zip file named Team-XX-A1.zip.
- Deadline for submission of solutions: **Sunday October 25th, 20:00 hrs**.
- Note that **if you have performed some parts of the assignment but do not discuss them in your report, you will not receive the associated points**. We will only evaluate your report; the code will only be consulted for verification purposes.
- Be concise and to the point.
- Use correct terminology and refer to scientific literature where appropriate.

Questions regarding the assignment can be asked through Canvas or during the sessions on **Thursday mornings**.

Good luck!

## Student Assignment 1

You are working as a data scientist for an e-commerce company that wants to improve its **customer feedback analysis**. Your task is to explore methods from **text representation, feature selection, and modeling** to understand product reviews and predict review sentiment.

Your manager provides you with the **Amazon Reviews dataset** on **CDs & Vinyl**, which contains product reviews along with their star ratings. You decide to transform this into a **binary classification task**:

- Label reviews with rating  $\geq 4$  as **positive (1)**.
- Label reviews with rating  $\leq 3$  as **negative (0)**.

In addition to classification, your manager is also curious about **clustering and topic modeling** to understand major themes in customer feedback.

You have **four weeks** to build models, compare approaches, and summarize your findings in a concise report.

### The dataset

- You will use a subset of the **CDs & Vinyl** category of the Amazon Reviews 2023 collection. A download link is provided on Canvas. For more information on the source of this dataset, see <https://amazon-reviews-2023.github.io/>
- Each review is a row in the file. The review body you need is **text** and the star rating is **rating**; asin, parent\_asin, user\_id, timestamp, helpful\_vote and verified\_purchase are also available.

### 1. Dataset Creation

- You change the data to fit a binary classification task, where the target variable is the rating that is associated with the review. You try to predict the rating based on the content of the review. Define your target label.
- What balance exists between positive and negative labels? Are there more negative or positive reviews?

### 2. Preprocessing & Analysis

- Tokenize and preprocess your dataset. Justify the choices you make.
- Should all reviews be included in the analysis? Again, justify your choices.
- Provide descriptive statistics, such as but not limited to:
  - Number of reviews
  - Average words per review
  - Number of unique words
- Show the **50 most common words** in the corpus after preprocessing (that is, before the stopwords removal in Task 3).

### 3. Bag-of-Words Model

- Remove stopwords. You can use the built-in stop word list from scikit-learn.
- Show the 100 most common words after stopwords removal.

- Calculate mutual information for these 100 words with the target label and report the most informative ones, and your interpretation.

#### 4. Feature Representation

- Generate **TF-IDF** vectors from the 100 most common words.
- Train a **Logistic Regression** model on the TF-IDF vectors such that each word is a feature and report the results.
- Train and analyze **Word2Vec models** (train your own vs. pretrained) on the original text data. Discuss pros/cons of each. Choose which one you prefer for this project.
- Explain how you convert word embeddings into document embeddings.

#### 5. Feature Selection & Dimensionality Reduction

- Apply **PCA (Principal Component Analysis)** and **ICA (Independent Component Analysis)** on your word2vec **word** embeddings to convert them to 2-dimensional space. Inspect and discuss whether similar words lie close to each other in 2-dimensional space.

#### 6. Modeling & Evaluation

- Train a **KNN classifier**, **Naïve Bayes**, **Logistic Regression** on your Word2Vec document representations. Note that multinomial Naïve Bayes requires non-negative features and so cannot be applied to embeddings; choose an appropriate variant and justify your choice.
- Implement **K-Fold Cross-Validation** to evaluate your models.
- Apply a small **hyperparameter optimization** on your Logistic Regression (e.g., GridSearchCV or RandomizedSearchCV). Be careful to distinguish the score the search itself reports from the performance you claim for your final model.
- Save and load your best-performing model, and include it in your submission (see the deliverables on page 1).
- Compare your different Word2Vec classifiers.

#### 7. Clustering & Topic Modeling

- Apply **two clustering algorithms** — Agglomerative Clustering and DBSCAN — on your Word2Vec **word** embeddings.
- Apply **Topic Modeling algorithms** — BERTopic and FuzzyTM — to extract topics from the reviews.
- Discuss the quality of the found topics, using appropriate metrics.

*Two practical notes on this task:*

- *BERTopic downloads a sentence-transformer model on first use. It also accepts precomputed embeddings, so you may supply a representation of your own instead.*
- *FuzzyTM builds dense vocabulary × document matrices and will exhaust an ordinary laptop's memory on an unpruned review vocabulary. Remove rare words before fitting it, and report the threshold you used.*

#### 8. Evaluation & Discussion

- Discuss which models and representations performed best and why.
- Reflect on the **limitations** of your approach and possible improvements.

## Report Guidelines

Prepare a report for your manager in which you explain the **modeling process, outcomes, and applicability** of your approach. Show in your report that you have **performed all of the tasks mentioned above**.

- Maximum length: **6 A4 pages** (excluding figures, tables, references and the technology statement).
- Include at least:
  1. Dataset creation
  2. Preprocessing
  3. Analysis
  4. Feature representation
  5. Feature selection and dimensionality reduction
  6. Training procedure
  7. Evaluation
  8. Clustering & topic modeling
  9. Discussion & limitations

These nine report sections do not map one-to-one onto the eight tasks. Task 2 supplies both “Preprocessing” and “Analysis”, and the results you produce in Task 6 belong under “Training procedure” and “Evaluation”. Use the section list above as the structure of your report, not the task numbering.

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You realize four weeks will fly by quickly — start early!

Good luck!